



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 4.10.24

Instructions:

Attempt all questions. The intended marks for questions are given in brackets []. This paper consists of four printed pages.

Question 1

Choose the correct answer from the choices given:

[6×1=6]

i) The trim() method of the String class removes _____.

- a) Leading spaces only b) trailing spaces only
- c) Spaces in between words d) both (a) and (b)

ii) Which one of the given statements is true for the following statement?

string1.compareTo(string2)

- a) if string1 > string2 the result will be a negative integer i.e. < 0.
- b) if string1 > string2 the result will be a positive integer i.e. > 0.
- c) if string1 > string2 the result will be 0.
- d) None of the above

iii) Assertion(A): In bubble sorting, the total number of iterations is equal to $1+2+3+....+n-1$, where n is the number of elements in the array.

Reason(R): Because the array is sequentially scanned multiple times and during each iteration, the pair of the consecutive elements are compared and interchanged

- a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
- b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
- c) Assertion (A) is true and Reason (R) is false
- d) Assertion (A) is false and Reason(R) is true

iv) Predict the output:

```
class Floyd
{ public static void main()
{   int notes[ ] = {1000, 500, 100, 50, 20, 10, 5, 2, 1 };
    int i, n, amt = 8888;
    for( i=3; i<5; i++)
    { n = amt/notes[i];
      if(n!=10)
        System.out.println("The number of notes " + notes[i] + " = " + n);
      amt = amt % notes[i];   }   } }
```

- | | |
|---------------------------------|---------------------------------|
| a) The number of notes 50 = 177 | b) The number of notes 1000 = 8 |
| The number of notes 20 = 1 | The number of notes 500 = 1 |
| c) The number of notes 1 = 8 | d) Error |
| The number of notes 2 = 5 | |

v) Predict the output:

```
class Flow
{ public static void main()
{   char a[ ]={'a', 'b', 'c', '@', '3'};
    String s=String.valueOf(a);
    System.out.print(s);   } }
```

- | | | | |
|--------------|----------|--------|------------------|
| a) No output | b) abc@3 | c) abc | d) Runtime error |
|--------------|----------|--------|------------------|

vi) A class can have any number of constructors but with different sets of _____.

- | | | | |
|------------|---------------|----------|------------|
| a) objects | b) parameters | c) names | d) members |
|------------|---------------|----------|------------|

Question 2

- a) Find the errors in the given code segment and write the statements correctly to assign values to an integer array. [2]

```
int a = new int(5);
for( int i = 0; i >= 5; i++)
i = a[i];
```

b) Predict the output: [2]

```
String c[] = {"DEHRADUN", "NAINITAL", "SHIMLA", "KASHMIR", "RANIKHET"};  
System.out.println(c[0].length()>c[3].length());  
System.out.println(c[4].substring(0,4));
```

c) Predict the output of the following: [5]

- i) "VIRTUAL BRAINPOWER".indexOf('e') + "x".concat("y")
- ii) (int)"Reliability".charAt(3) + 'A'
- iii) "VERSATILITY".substring(-2,4).concat("PROGRAM")
- iv) "TECHnoSoft".startsWith("Micro") ? "right" : "wrong"
- v) "IndiaWins".compareToIgnoreCase("Indiawins12")

d) Difference between the following statements: [2]

```
object obj = new object( );  
object obj = new object(4,5,7);
```

e) If x[] = { -1, 0, 15, -2, 6}; What are the values of p, q and r ? [3]

```
p = x.length + x[2]  
q = x[2] * x[1] - x[4]++  
r = ++x[1] - --x[3] + x[3];
```

Question 3 [10]

Write a program to input a sentence and print each word along with its length in tabular form.

For example: Input: I study in Delhi Public School Newtown

Output:

Word	Length
I	1
study	5
in	2
Delhi	5
Public	6
School	6
Newtown	7

Question 4

[10]

Design a class to overload a function Joy1() as follows:

1. void Joy1(String, char, char) : The function replaces the first character argument with the second character argument from the given string and prints the new string.
2. void Joy1(String) : The function prints the position of the first space and the position of the last space from the given string .
3. void Joy1(String, String) : The function combines the two strings with a space between them and prints the resultant string.

Do not write the main method